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# FOOD DELIVERY MANAGEMENT SYSTEM

## DBMS ASSIGNMENT

Customers , Restaurants, Orders ,  
Payments, Delivery



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# NAME AND PURPOSE OF THE APPLICATION

## Application Name:

Food Delivery Management System

## Purpose:

A Food Delivery Management System is used to manage online food ordering and delivery. It connects Customers, Restaurants, Food items, Orders, Payments, and Delivery partners in one database.

### **The database helps to:**

- Store customer information
- Manage restaurants and their menus
- Record customer orders
- Track payments
- Assign delivery partners
- Maintain delivery status

# REQUIRED TABLES

*A practical Food Delivery Management System can be represented using 7 tables:*

|                         |
|-------------------------|
| <b>Customer</b>         |
| <b>Restaurant</b>       |
| <b>Menu_Item</b>        |
| <b>Orders</b>           |
| <b>Order_Item</b>       |
| <b>Payment</b>          |
| <b>Delivery_Partner</b> |

# TABLES AND THEIR ATTRIBUTES

## Table 1: CUSTOMER

| Column        | Data Type | Key         |
|---------------|-----------|-------------|
| Customer_ID   | INT       | Primary Key |
| Customer_Name | VARCHAR   | -           |
| Phone         | VARCHAR   | -           |
| Email         | VARCHAR   | -           |
| Address       | VARCHAR   | -           |

### Purpose:

Stores the details of customers who use the food delivery application.

## Table 2: RESTAURANT

| Column          | Data Type | Key         |
|-----------------|-----------|-------------|
| Restaurant_ID   | INT       | Primary Key |
| Restaurant_Name | VARCHAR   | -           |
| Phone           | VARCHAR   | -           |
| Rating          | DECIMAL   | -           |
| Address         | VARCHAR   | -           |

### Purpose:

Add a little bit of body text about restaurants available on the application

## Table 3: MENU\_ITEM

| Column          | Data Type | Key         |
|-----------------|-----------|-------------|
| Item_ID         | INT       | Primary Key |
| Restaurant_Name | VARCHAR   | Foreign Key |
| Item_Name       | VARCHAR   | -           |
| Category        | VARCHAR   | -           |
| Price           | DECIMAL   | -           |
| Availability    | BOOLEAN   | -           |

### Purpose:

Stores the food items offered by each restaurant.

### Relationship:

One restaurant can have many menu items.

## Table 4: ORDERS

| Column        | Data Type | Key         |
|---------------|-----------|-------------|
| Order_ID      | INT       | Primary Key |
| Customer_ID   | INT       | Foreign Key |
| Restaurant_ID | INT       | Foreign Key |
| Order_Date    | DATE      | -           |
| Total_Amount  | DECIMAL   | -           |
| Order_Status  | VARCHAR   | -           |

### Purpose:

Stores information about orders placed by customers.

### Relationship:

- A customer can place many orders.
- A restaurant can receive many orders.

## Table 5: ORDER\_ITEM

| Column        | Data Type | Key         |
|---------------|-----------|-------------|
| Order_Item_ID | INT       | Primary Key |
| Order_ID      | INT       | Foreign Key |
| Item_ID       | INT       | Foreign Key |
| Quantity      | INT       | -           |
| Subtotal      | DECIMAL   | -           |

### Purpose:

Stores the individual food items included in each order..

### Relationship:

- One order can contain many order items.
- One menu item can appear in many orders.

## Table 6: PAYMENT

| Column         | Data Type | Key         |
|----------------|-----------|-------------|
| Payment_ID     | INT       | Primary Key |
| Order_ID       | INT       | Foreign Key |
| Payment_Date   | DATE      | -           |
| Payment_Method | VARCHAR   | -           |
| Amount         | DECIMAL   | -           |
| Payment_Status | VARCHAR   | -           |

### Purpose:

Records payment information associated with each order.

### Relationship:

An order is associated with a payment record.

# Table 7: DELIVERY\_PARTNER

| Column          | Data Type | Key         |
|-----------------|-----------|-------------|
| Delivery_ID     | INT       | Primary Key |
| Order_ID        | INT       | Foreign Key |
| Partner_Name    | VARCHAR   | -           |
| Phone           | VARCHAR   | -           |
| Delivery_Status | VARCHAR   | -           |

## Purpose:

Stores information about the delivery partner responsible for delivering an order.

## Relationship:

A delivery partner can deliver multiple orders over time.

# RELATIONS BETWEEN THE TABLES

| Relationship              | Cardinality | Meaning                                   |
|---------------------------|-------------|-------------------------------------------|
| Customer → Orders         | 1 : M       | One customer can place many orders        |
| Restaurant → Menu_Item    | 1 : M       | One restaurant can have many menu items   |
| Restaurant → Orders       | 1 : M       | One restaurant can receive many orders    |
| Orders → Orders_Item      | 1 : M       | One order can contain multiple items      |
| Menu_Item → Orders_Item   | 1 : M       | One menu item can occur in many orders    |
| Orders → Payment          | 1 : 1       | Each order has one payment record         |
| Orders → Delivery_Partner | 1 : 1       | An order is assigned to a delivery record |